

HPS-GPR v5.1.1

Significance-field covariance and signed signal response

Support study for analysis note v5.0.2

10 September 2026

This bounded pilot develops the covariance study in v5.1.0 and supplies new examples for Section 6.11 of the v5.0.2 note. It extends the conditional 2015 field through 100 MeV, resolves a finer 2016 covariance grid around the problematic 76 MeV region, and tests what the fit does to known positive signals. Negative fitted responses are retained throughout. The output is a diagnostic of specified background models, not calibrated global discovery significance or an extension of the released observed search.

Two covariance objects. The background GP predicts bin counts and their covariance inside a masked window. The significance field describes how the signed fitted root r_m varies *across tested masses* when one complete background spectrum fluctuates. For a coherent mean B ,

$$a_m = r(B; m), \quad D_{im} = r(B + \sqrt{B_i} e_i; m) - a_m, \quad C = D^T D, \quad s_m = \sqrt{C_{mm}}, \quad R_{mn} = C_{mn}/(s_m s_n).$$

Both signs of the unconstrained fitted signal enter r . The response matrix has no division by the number of support bins. Its columns are evaluated on the same input-bin basis; their inner products generate both positive and negative correlations. Centering this conditional field does not make a biased reference into a qualified discovery null.

Campaign	Nodes	Added	RMS $R_{\text{emp}} - R$	Rank metric	Median CI width
2015	82	10	0.056	7.9	0.228
2016	153	11	0.063	9.3	0.222
2021	201	0	0.059	15.8	0.234

The rank metric is $(\text{tr} R)^2 / \text{tr}(R^2)$, a description of correlated modes; it is not an effective trials factor. The final column summarizes pointwise bootstrap intervals for the predeclared correlation slices, not uncertainty in the physical background.

Bounded design. All integer nodes are retained: 2015 at 19–100, 2016 at 39–180 and 2021 at 50–250 MeV. Eleven half-integer nodes from 70.5 to 80.5 MeV supplement 2016. The 2015 kernel is clamped to its archived 90 MeV state above 90; positive kernel parameters are interpolated in log space at the 2016 half-integers. Count supports and binning are unchanged. The same first 256 of the archived 1,000 complete Poisson spectra are used at every coordinate, including new fits at all added nodes. The initial wider fine-grid plan was reduced after timing the upper-end fits; this resource decision did not use their response values. Both protocols are saved.

Significance-field covariance: complete scans and independent Poisson checks

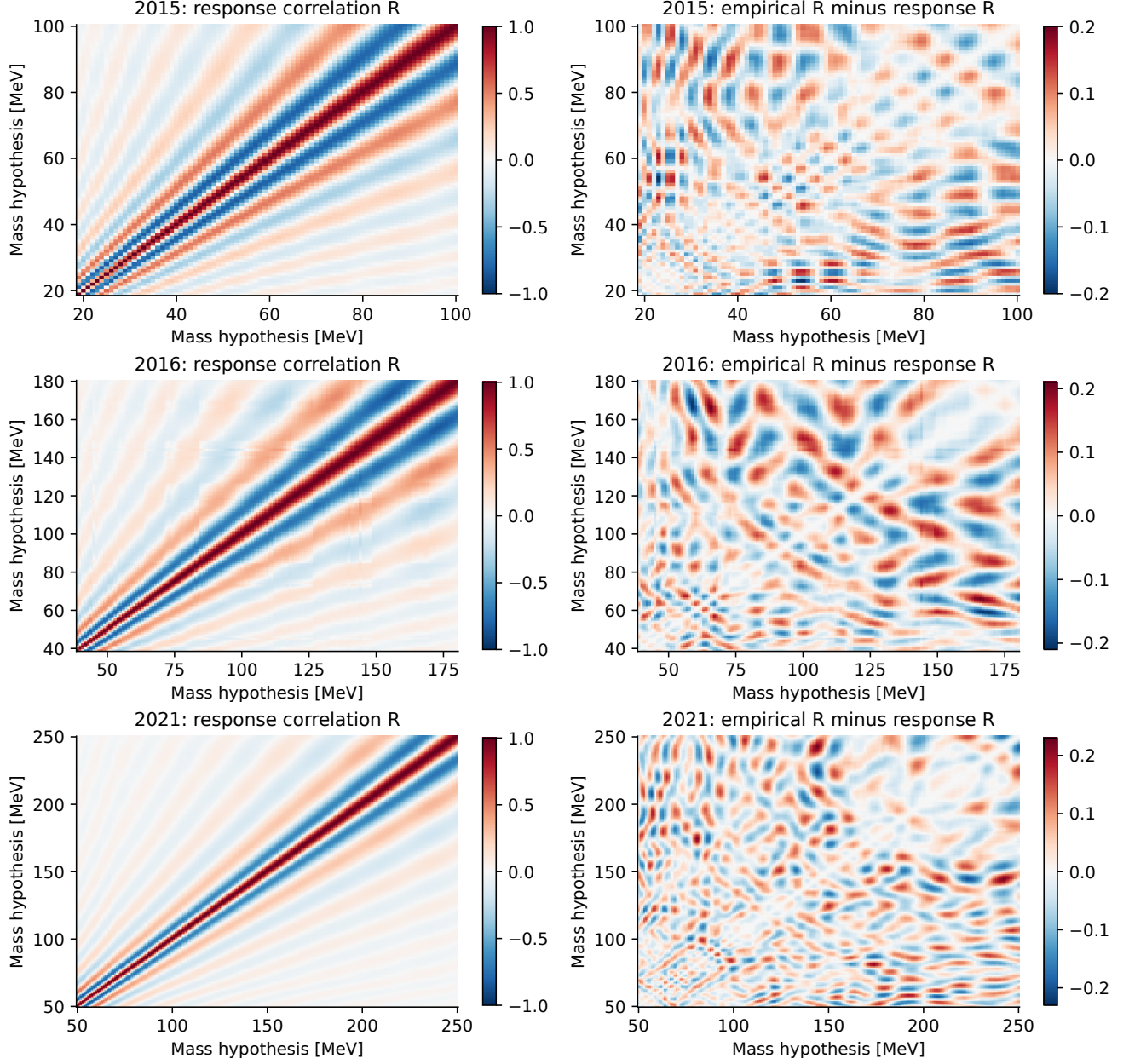


Figure 1: Response correlations and empirical differences from 256 coherent complete Poisson scans. Each matrix uses its actual irregular mass grid. Color scales for the differences are shown separately. The original 1,000-scan native-grid results remain an archived benchmark; they are not spliced into the extended 256-scan empirical matrix. Covariance agreement is a central-distribution check, not validation of a rare global tail.

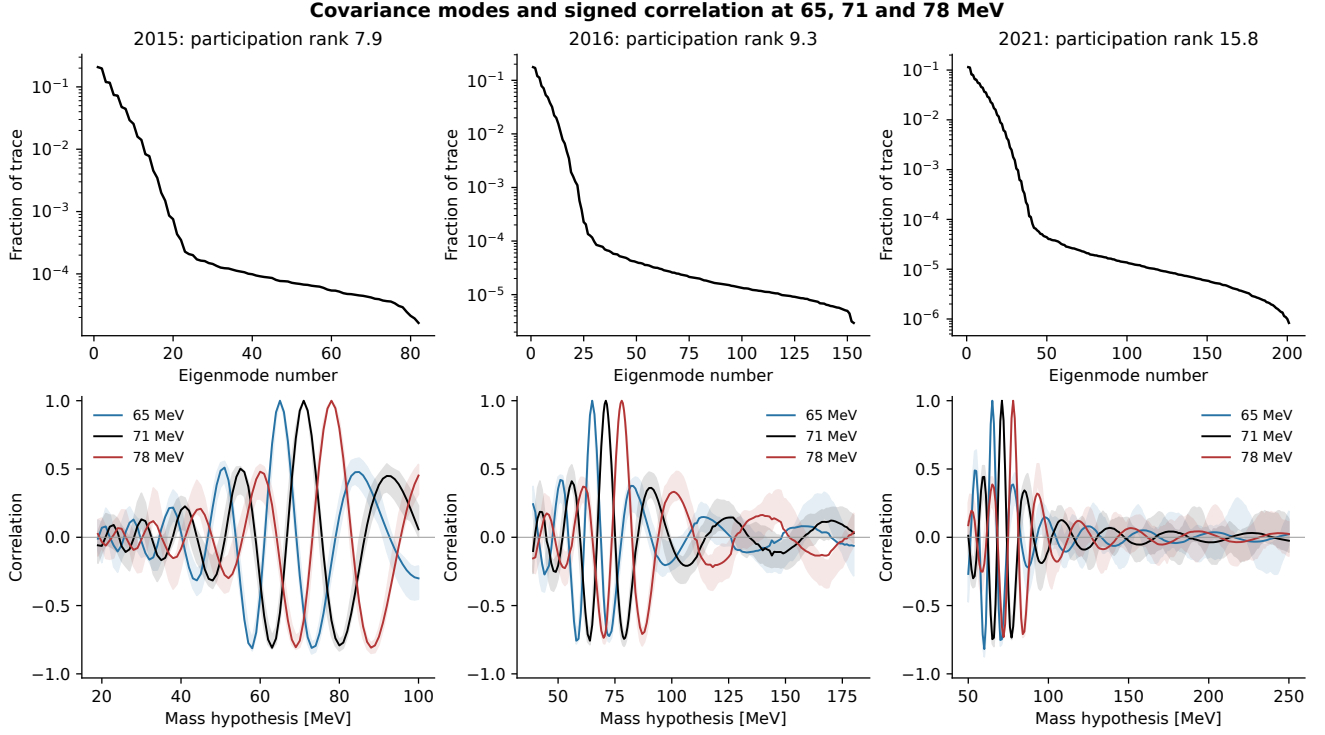


Figure 2: Eigenvalue fractions and signed correlations at 65, 71 and 78 MeV. Lines are the response prediction; shading is the pointwise 95% percentile interval from 128 resamples of complete Poisson experiments. Bands need not cover the line where the linear approximation differs from the empirical field. These intervals are neither simultaneous nor a systematic-error band.

Campaign	Masses [MeV]	Response R	Empirical R	Bootstrap interval
2015	65, 71	-0.552	-0.522	[-0.615, -0.428]
2015	71, 78	-0.608	-0.642	[-0.722, -0.553]
2015	65, 78	-0.187	-0.158	[-0.292, -0.031]
2016	65, 71	-0.643	-0.605	[-0.668, -0.538]
2016	71, 78	-0.682	-0.703	[-0.760, -0.616]
2016	65, 78	0.003	-0.004	[-0.122, 0.118]
2021	65, 71	-0.746	-0.733	[-0.780, -0.648]
2021	71, 78	-0.655	-0.643	[-0.698, -0.563]
2021	65, 78	0.387	0.362	[0.223, 0.442]

The fixed 65/71/78 MeV comparison connects to v5.2.0, which showed substantial negative correlations of the middle coordinate with its neighbors. Its observed conditional residual was a post-selection diagnostic. Here the same locations are used to explain the covariance structure; no probability for discovering that pattern is assigned.

Fixed-mass signal injection: response and background offset

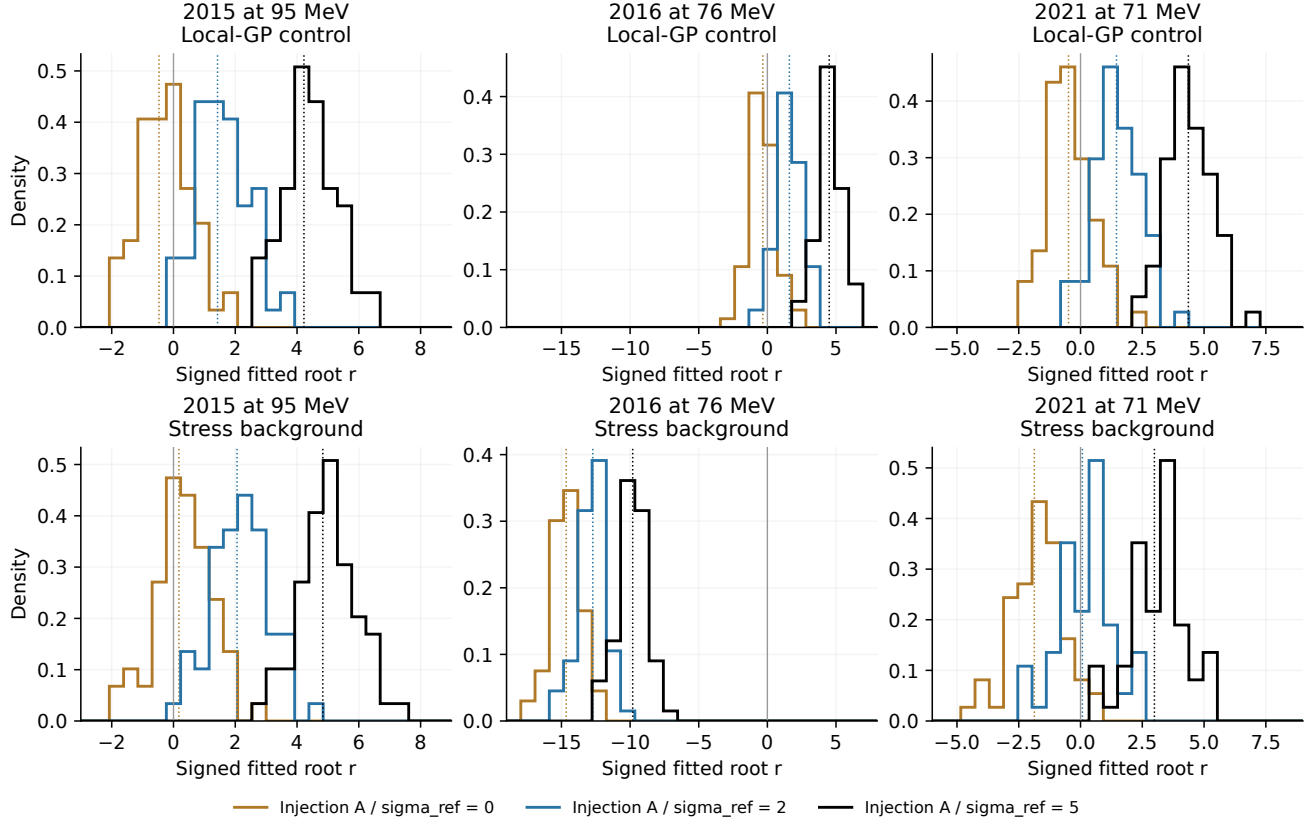


Figure 3: Fixed-mass injection examples. Each curve uses 64 Poisson experiments, paired across injected strengths within a background choice. Dotted lines show fits to unfluctuated expectations. Injection labels $A/\sigma_{\text{ref}} = 0, 2, 5$ specify yields, not achieved significances. The same fixed reference uncertainty is used for the two truths at a given mass. Both excesses and deficits are visible.

Campaign, mass	Truth	Mean r_0	Mean r_2	Mean r_5	$r_5 > 0$
2015, 95	Local GP	-0.27	1.62	4.41	64/64
2015, 95	Stress	0.33	2.19	4.98	64/64
2016, 76	Local GP	-0.32	1.62	4.52	64/64
2016, 76	Stress	-14.64	-12.70	-9.80	0/64
2021, 71	Local GP	-0.47	1.46	4.37	64/64
2021, 71	Stress	-1.72	0.22	3.14	64/64

The generating control is the full-support local-GP mean with the injected mass excluded from training. It is conditional on the observed sidebands and is different at each fixed injection mass; these control experiments are not combined into a global scan. The stress truth is the archived coherent full-spectrum reference. Background counts are Poisson draws; independent positive signal increments of size two and then three reference uncertainties give the paired size-two and size-five spectra. A more positive fit after injection demonstrates response, while any persistent offset exposes bias under that specified generating shape. A positive-fit fraction alone is not a discovery efficiency at a calibrated false-positive rate.

A positive injected signal can create neighboring negative fit responses

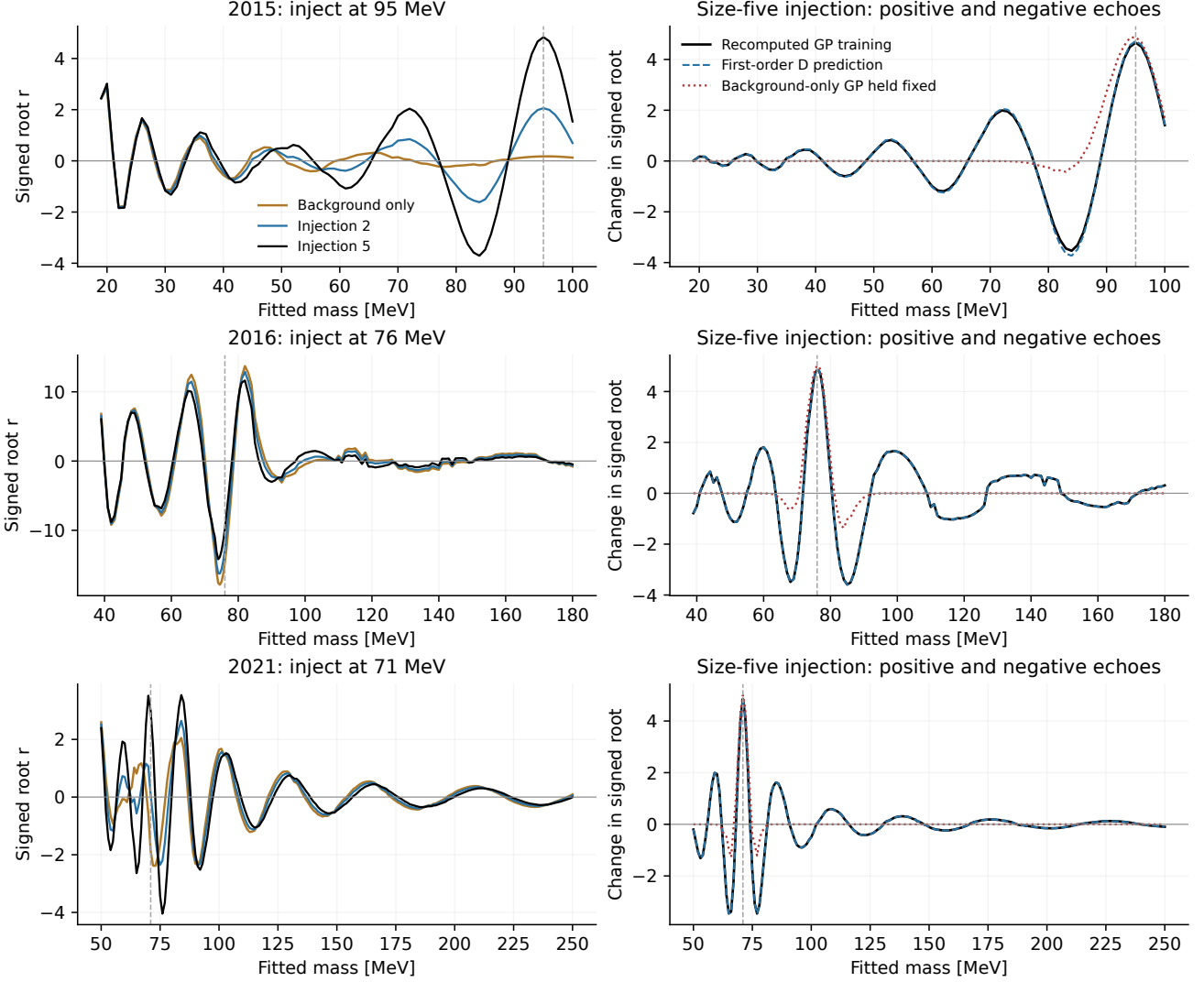


Figure 4: Whole-spectrum deterministic injections into the stress background. Left: raw signed roots before and after injection. Right: the size-five change with full retraining, its first-order prediction, and a control holding the background-only GP prediction/covariance fixed. The likelihood background nuisance parameters are still profiled in the last control. Vertical lines mark the injected mass. Negative neighboring responses are retained, not converted to zero.

Campaign, injection	Δr at injection	Minimum echo	Echo mass	Linear RMSE
2015, 95	4.659	-3.536	84	0.059
2016, 76	4.843	-3.589	85	0.002
2021, 71	4.860	-3.457	77	0.001

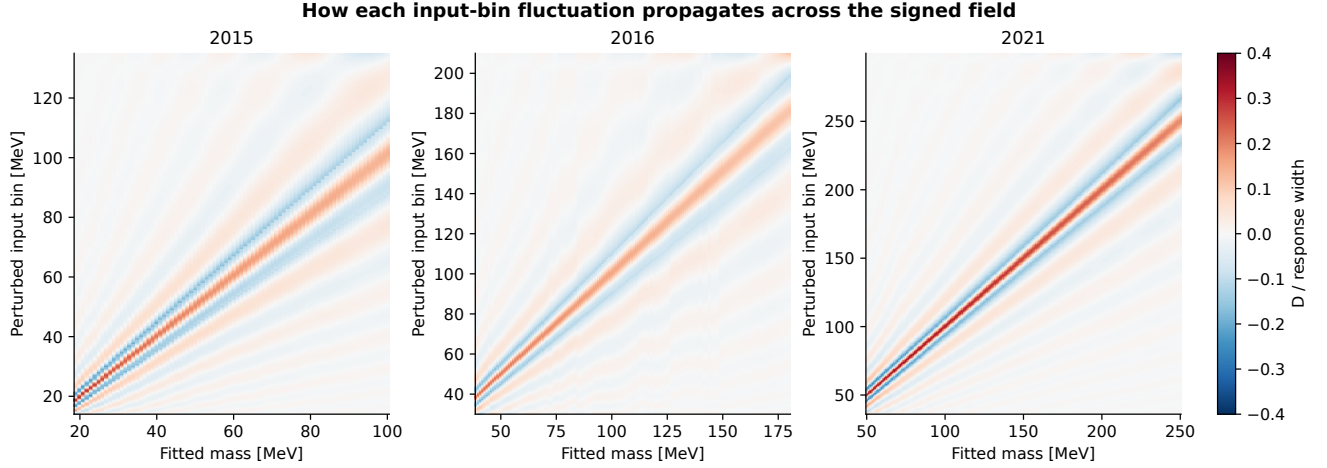


Figure 5: Each column shows the signed response to independent input-bin perturbations, divided by that fitted mass’s response width. Positive and negative regions encode the moving-window interpolation. All values are retained in the matrices; color saturation is a display choice.

For a known injected spectrum δn , the response predicts

$$\Delta r_m \simeq \sum_i D_{im} \frac{\delta n_i}{\sqrt{B_i}}.$$

This is a first-order prediction of the fitting procedure, not an identity between a signal template and a covariance column. Finite injections can change the count-dependent GP errors, its mean and covariance, and the profiled likelihood. The discrepancy in the preceding table measures that nonlinearity. The fixed-training control separates the consequences of retraining from the fit’s response with a fixed GP constraint.

Connection to v5.2.1–2. The earlier moving-mask pair/triplet injections and frozen-training controls motivate retaining the complete signed pattern. Their observed-fit amplitudes and family choices are not used as new null parameters here. The v5.2.2 weak-middle study also showed that a limit uplift need not fill a negative trough: a middle root can already be positive in some campaigns. For that reason this pilot reports signed roots and their changes directly, with no new family ranking.

Limits of this study. The 2016 reference is still not independently qualified. Its large unfluctuated root measures how that reference is represented by the estimator, not a measured resonance. The 2015 extension is conditional on a frozen endpoint kernel, and the denser 2016 grid is a numerical diagnostic. No new local or global significance is claimed, no scan maximum is calibrated, and no released limit or observed p-value is changed. Complete-scan injected-signal power and independent background qualification remain separate future tests.

Reproducibility. The fixed masses are 2015: 51, 78, 95 MeV; 2016: 42, 66, 76, 92 MeV; 2021: 71, 92 MeV. The protocols, seed convention, all 3,456 fixed-mass scenario rows, original input hashes, covariance checkpoints, deterministic scan values, bootstrap slices and numerical checks are bundled. One worker and one linear-algebra thread were used. The source archive regenerates figures and this report from saved inputs; refitting additionally requires the recorded HPS runtime and original data.